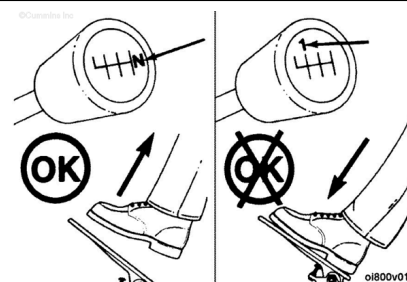


Trainings ▾

General Information

⚠ CAUTION ⚠

To reduce the possibility of damage to the starter, do not engage the starting motor for more than 30 seconds. Wait 2 minutes between each attempt to start the engine (electrical starting motors only).



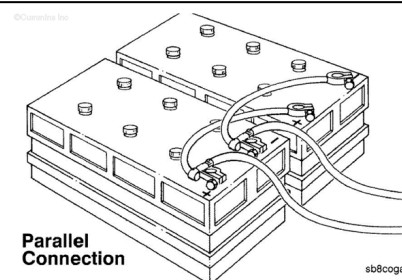
Engines equipped with air starters require a minimum of 480 kPa [70 psi] compressed air pressure.

Disengage the driven unit, or, if equipped, put the transmission in neutral.

Start the engine with the throttle in the idle position.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



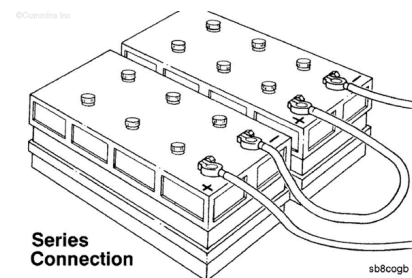
⚠ CAUTION ⚠

When using jumper cables to start the engine, make sure to connect the cables in parallel: positive (+) to positive (+) and negative (-) to negative (-). When using an external electrical source to start the engine, turn the disconnect switch to the OFF position. Remove the key before attaching the jumper cables.

The illustration shows a typical parallel battery connection. This arrangement doubles the cranking amperage.

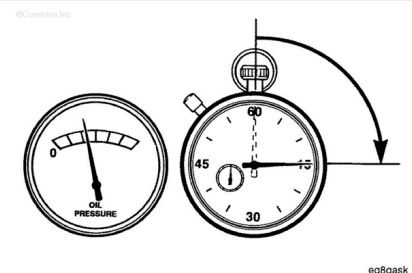
The illustration shows a typical series battery connection. With this arrangement the positive (+) battery cable is connected to the adjoining negative (-) battery terminal.

This arrangement doubles the voltage.



⚠ CAUTION ⚠

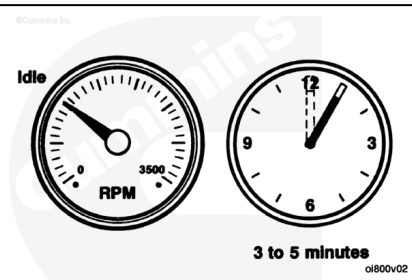
Engine oil pressure must be indicated on the gauge within 15 seconds after starting. If oil pressure is not registered within 15 seconds, shut off the engine immediately to avoid engine damage.



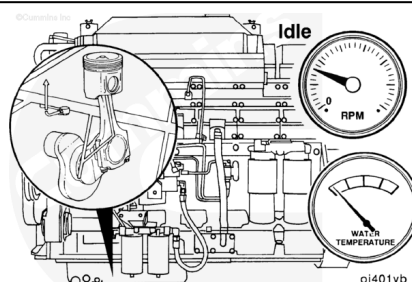
Confirm the correct oil level in the oil pan.

For engines equipped with a Prelub™ system, the starter will **not** engage until adequate oil pressure is obtained.

Idle the engine 3 to 5 minutes at approximately 1000 rpm, and attain a minimum water temperature of 60°C [140°F] before operating with a load.

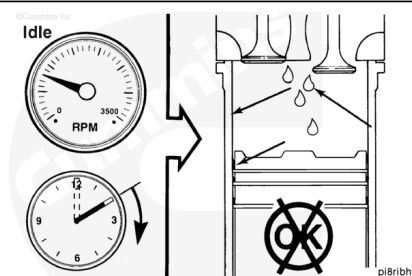


Increase engine speed (rpm) slowly to provide adequate lubrication to the bearings, and allow the oil pressure to stabilize.



⚠ CAUTION ⚠

Do not idle the engine for excessively long period. Long periods of idling, more than 10 minutes, can damage an engine because combustion chamber temperatures drop so low the fuel will not burn completely. This will cause carbon to clog the injector spray holes and piston rings and can cause valves to stick.



⚠ CAUTION ⚠

If the engine coolant temperature becomes too low, 60°C [140°F], raw fuel from incomplete combustion, will wash the lubricating oil off the cylinder walls and dilute the crankcase oil; therefore, all moving parts of the engine will not receive the correct amount of lubrication.

